Ben Mackert

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Education

University of Colorado Boulder, Boulder, CO
M.S. Electrical and Computer Engineering — Expected May 2028
B.S. Electrical and Computer Engineering — Expected May 2027
GPA 3.9, Dean's List

Technical Courses

Probability Theory (A), Linear Systems (A), Data Structures (A), Computer Organization (A), Embedded Software Eng. (A), RTOS (A–), Circuits I & II (A), Electronics Design Lab (A–), Prog. Digital Systems (A–), Calculus I–III

Skills

Languages: C, C++, Python, RISC-V Assembly, Verilog, LaTeX, Java, HTML, CSS

Software: ArduPilot SITL, MAVLink, pytest, OpenCV, MediaPipe, Altium Designer, SolidWorks, LTspice, STM32CubeIDE, GitHub, Bitbucket, Excel

Core Engineering Skills: Embedded Systems, Computer Vision, Machine Learning, SLAM & State Estimation, Embedded Firmware, Circuit Design, Digital Logic, Sensor Interfaces, Data Logging

Hardware & Testing: PCB Design, Hardware Bring-Up, Sensor Integration, SMD, THT, Soldering, Oscilloscopes, Function Generators, DMMs, Power Supplies

Experience

Co-Founder & CTO – Skopos Industries Inc. **San Francisco, CA** — Jun 2026 – Present

- Co-founded a defense startup building a self-flying drone that maps GPS-denied buildings onto AR glasses; lead all software and serve as SBIR principal investigator.
- Architected ESP32 firmware, a BLE protocol, a Python mapping pipeline and AR display, backed by 1,600+ automated tests.
- Built a lidar scan-matching pose estimator reaching 3.1 cm average wall error in simulation, and added an observability check that rejects confident-but-wrong fixes that drifted 13.3 m along featureless walls.
- Flew scan-matched pose on ArduCopter SITL over MAVLink with no GPS (hover within 9 cm); fault injection exposed flyaways the autopilot's health checks missed, now gates for first flight.
- Designed for safe failure: stale sensor data shows as unknown, never clear, and the system never arms the drone itself.

Electrical & Software Engineering Intern – Medtronic **Boston, MA** — Jun 2026 – Aug 2026

- Developed a surgeon head-pose tracking pipeline using Intel RealSense, OpenCV, MediaPipe Face Mesh, and solvePnP to analyze RGB/NIR camera tracking against a motion-capture reference system.
- Synchronized motion-capture head-pose data with camera frame timestamps over DDS for post-session and real-time comparison of stereo NIR, RGB, and reference tracking solutions.
- Built a Python GUI to manage capture metadata, log synchronized tracking data, and streamline computer vision analysis across test sessions.
- Supported STM32 board bring-up for a surgical robotics prototype, including debugging hardware issues, and developing I2C firmware for sensor data acquisition.

Electrical Engineering Intern – CONMED Corporation **Denver, CO** — May 2025 – Aug 2025

- Designed and built an auto-calibration system using an Arduino Nano Every and digital potentiometer to map ESU voltages to resistance with a 4^{th}-degree polynomial.
- Designed a custom PCB in Altium for the calibration hardware, integrating a resistor ladder with MCU control.
- Created a Python Tkinter GUI and serial/Matplotlib backend to display real-time calibration plots.
- Reduced calibration error by 92.5%, enabling full transformer compatibility and cutting manufacturing scrap.
- Simulated 3- and 4-point calibration in Python; achieved LMS errors as low as 1.76 Ω^2 across transformer models.

Projects

STM32 Microcontroller PCB – Bluetooth & USB-C

- Created a 4-layer STM32 PCB in Altium Designer for Bluetooth and USB-C integration.
- Achieved 480 Mbps USB-C data and BLE telemetry for wireless diagnostics between devices in testing scenarios.
- Optimized high-speed signal routing against EMI by collaborating with 3 engineers on layout constraints and PCB stack-up.
- Engineered oscilloscope-based test methods, boosting debugging efficiency by 20% and accelerating bring-up.